Useful Resources

Self-hosted inference and Option 2 lab hardware

- llama.cpp: a candidate local inference runtime; verify support and performance for the actual model and device.
- OpenVINO GenAI NPU documentation: a starting point for supported Intel NPU experiments in Option 2.
- AMD ROCm compatibility matrix: check the installed operating system, driver, hardware, and runtime combination for Option 2 lab experiments.
- GMKtec EVO-X2 and ASUS NUC 16 Pro: product-level information for Option 2 lab machines; course-machine configuration will be confirmed separately.

Agent security and evaluation

- AgentDojo: prompt-injection evaluation for tool-using agents.
- TAMAS: adversarial multi-agent scenarios.
- A2ASecBench: protocol-aware multi-agent security evaluation.
- NYU CTF Bench: containerized cybersecurity challenges for evaluating agents.

Fuzzing

- OSS-Fuzz-Gen: LLM-assisted fuzz-target generation and evaluation.
- OSS-Fuzz documentation: build, harness, and evaluation guidance.
- LLVM libFuzzer: coverage-guided fuzzing and execution controls.

AWS Resources

Each student has $50 in AWS credits with a $3 daily spending limit; credits stack within a team. Access will be enabled after the team formation deadline and announced on Piazza.

Available services

- EC2: virtual machines for inference endpoints, agents, and experiments.
- EBS: disk volumes attached to EC2 instances (OS, code, model weights); they persist when an instance is stopped and are billed by size even when it is not running.
- S3: object storage for datasets, model files, logs, and results, accessible from any instance.

If your team needs other AWS services, let the teaching team know as soon as possible so we can request access from the university.

EC2 instances with GPU (GPU memory, on-demand price)

- g6f.large: 1/8 NVIDIA L4 (3 GB), $0.20/hr
- g6f.xlarge: 1/8 NVIDIA L4 (3 GB), $0.24/hr
- g6f.2xlarge: 1/4 NVIDIA L4 (6 GB), $0.48/hr
- g4dn.xlarge: NVIDIA T4 (16 GB), $0.53/hr
- g6.xlarge: NVIDIA L4 (24 GB), $0.81/hr
- g6.2xlarge: NVIDIA L4 (24 GB), $0.98/hr
- g5.xlarge: NVIDIA A10G (24 GB), $1.01/hr
- g6e.xlarge: NVIDIA L40S (48 GB), $1.86/hr

CPU-only instance types are also available.

Practical experiment guidance

Use a pinned course-tested version of the chosen stack. Start with a pilot before launching a large experiment matrix. Save workload definitions, seeds, configuration, logs, and analysis code together. Preserve failed outcomes and explain exclusions.

For AWS experiments, confirm that the service is eligible for the university credit program before provisioning it. Set a project budget and shut down unused resources. Options 1 and 3–7 use AWS instances by default, including self-hosted CPU inference when paid API access is unavailable. Shared lab-machine inference is reserved for Option 2. **Available GPU instances are listed under AWS resources above; for LLM API usage, use TritonAI.**

Project deadlines and group policies are listed in the Overview tab. Course-specific access instructions, software images, Option 2 machine reservation procedures, AWS account instructions, and grading weights will be added when confirmed.